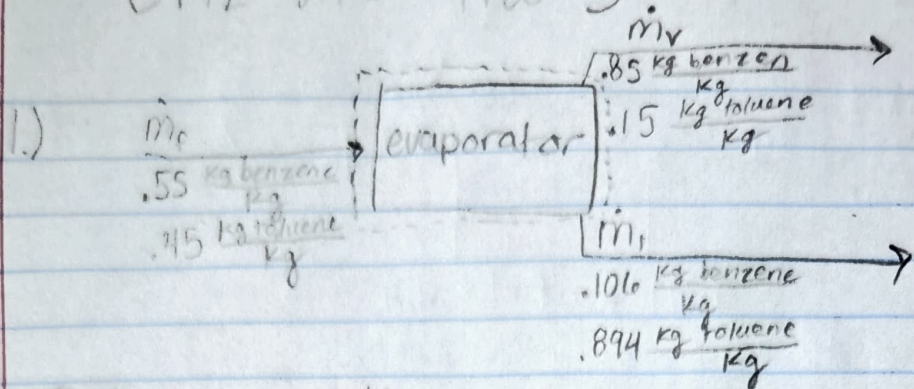


CHE 210 Hw 3



Input/output table

input			out.put			
species	mass %	$m_i (\frac{kg}{s})$	mass %	$m_i (\frac{kg}{s})$	mass %	$m_i (\frac{kg}{s})$
Benzene	55	$.55 m_F$	85	$.85 m_V$	10.6	$.106 m_L$
Toluene	45	$.45 m_F$	15	$.15 m_V$	89.4	$.894 m_L$
Total	100	m_F	100	m_V	100	m_L

Mass balance: $m_F = m_V + m_L$

Benzene: $X_{FB} m_F = X_{VB} m_V + X_{LB} m_L$ ✓

Toluene: $X_{FT} m_F = X_{VT} m_V + X_{LT} m_L$ ✓

Basis: $m_F = 1 \frac{kg}{s}$

$$m_V = 1 - m_L$$

$$.55 (1 \frac{kg}{s}) = .85 (1 - m_L) + .106 m_L$$

$$.55 = .85 - .744 m_L$$

$$.744 m_L = .3$$

$$m_L = .403 \frac{kg}{s}$$

$$m_V = 1 - .403 = .5968 \frac{kg}{s}$$

$$\frac{m_V}{m_L} = 1.48$$

DOF:

Composition of streams: 6

Chem rxn: 0

Accumulation: 0

specified flow rate: 0

specified composition: 3

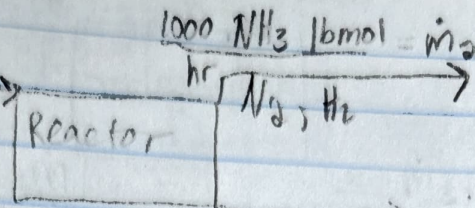
Binary system: 2

$6 - 5 = 1$ The problem is underspecified. ✓

2) Murphy 2.39

0.25 mol N_2
total mol
0.75 mol H_2
total mol

$\dot{n}_1$ H_2
 N_2



$T = 1440 \text{ R}$ $P = 100 \text{ atm}$

$$\frac{1000 \text{ lbmol } NH_3}{\text{hr}} \times \frac{1 \text{ lbmol } N_2}{2 \text{ lbmol } NH_3} = 500 \text{ lbmol } N_2 \text{ reacted} \quad \checkmark$$

$$\frac{500 \text{ lbmol } N_2}{\text{hr}} \times \frac{1 \text{ lbmol } N_2}{0.25 \text{ lbmol } N_2} = 2000 \text{ lbmol } N_2 \text{ in} \quad \checkmark$$

$$\frac{2000 \text{ lbmol } N_2}{\text{hr}} \times 3 \frac{\text{lbmol } H_2}{1 \text{ lbmol } N_2} = 6000 \text{ lbmol } H_2 + \frac{2000 \text{ lbmol } N_2}{\text{hr}} = \frac{8000 \text{ lbmol}}{\text{hr}} \quad \checkmark$$

$V = \frac{nRT}{P}$ $R = 10.73 \frac{\text{ft}^3 \text{ atm}}{\text{lbmol} \cdot \text{R}}$ 0.73024 $\checkmark$

$$V = \frac{8000 \text{ lbmol}}{\text{hr}} \times \frac{10.73 \text{ ft}^3 \cdot \text{atm}}{\text{lbmol} \cdot \text{R}} \times 1440 \text{ R} \times \frac{1}{100 \text{ atm}} \quad \checkmark$$

$= 1.24 \times 10^6 \frac{\text{ft}^3}{\text{hr}} \quad \times$

18
20

3.2.58 Broth: $\frac{.15 \text{ g glucose}}{\text{g total}}$, $\frac{.06 \text{ g phosphate}}{\text{g total}}$, $\frac{.06 \text{ g Nitrate}}{\text{g total}}$, $\frac{.73 \text{ g water}}{\text{g total}}$

a.) Matrix? System of eq

Basis: Making 1000g of Broth

Broth = Fast Feed(x) + Super gro(y) + Formula N(z) + Water = 1

$$G: 150 = 450x$$

$$P: 60 = 20x + 350y + 120z$$

$$N: 60 = 10x + 150y + 580z$$

$$\begin{bmatrix} 450 & 0 & 0 \\ 20 & 350 & 120 \\ 10 & 150 & 580 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 150 \\ 60 \\ 60 \end{bmatrix}$$

Matrix multiplication:

$$x = \frac{1}{3} \quad y = .1305 \quad z = .064$$

$$w = 1 - \left(\frac{1}{3} + .1305 + .064\right) = .47229 \frac{\text{kg water}}{\text{kg}}$$

$$\text{Fast Feed} = 1000 \text{ g tot} \times \frac{19x}{3 \text{ g tot}} = \frac{1000}{3} \text{ g needed}$$

$$\text{Super gro} = 1000 \text{ g tot} \times \frac{.1305 \text{ g y}}{\text{g tot}} = 130.5 \text{ g}$$

$$\text{Formula N} = 1000 \text{ g tot} \times \frac{.064 \text{ g z}}{\text{g tot}} = 64 \text{ g needed}$$

$$\text{Water} = 1000 \text{ g tot} \times \frac{.47229 \text{ w}}{\text{g tot}} = 472 \text{ g needed}$$

b.) Normal: $\frac{15 \text{ g}}{\text{kg}}$

Homemade:

$$\text{Fast Feed: } \frac{1000 \text{ g}}{3} \times \frac{20 \text{ g}}{1000 \text{ g}} = \frac{20}{3} = 6.67 \text{ g}$$

$$\text{Super gro: } 130.5 \text{ g} \times \frac{12}{1000 \text{ g}} = \$1.31$$

$$\text{Formula N: } 64 \text{ g} \times \frac{15}{1000 \text{ g}} = \$0.96$$

$$\text{Total} = \$6.67 + \$1.31 + \$0.96 = \$8.97$$

$$\text{Savings: } 15 - 8.97 = \$6.03$$

Need to save \$2000

$$\frac{\$6.03}{\text{kg}} \times 20 \text{ kg} = 120.6 \$ \text{ saved per batch}$$

$$\frac{\$2000}{\$120.6} \times \frac{1 \text{ batch}}{\$120.6} = 16.6 \text{ batch} \rightarrow 17 \text{ batches}$$

c.) Current trace: $\left(\frac{.03 \text{ g}}{\text{g x}} \cdot \frac{19 \text{ g}}{3} \right) \left(\frac{.08 \text{ g}}{\text{g y}} \cdot .1305 \text{ g y} \right) = \frac{.020 \text{ g}}{\text{g total}} = 2\% \text{ wt}$

$$x + y + z + w = 1$$

$$.45x = .15 \quad x = \frac{1}{3}$$

$$.02x + .35y + .12z = .06$$

$$.01x + .15y + .58z = .06$$

$$.03\left(\frac{1}{3}\right) + .08y = .03$$

$$y = .25$$

28.
30

$$.02\left(\frac{1}{3}\right) + .35(.25) + .122 = .06$$

$$.122 = .034167$$

$$z = .285 \text{ (impossible)} \quad \times$$

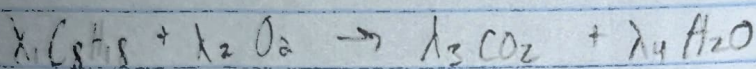
This, we are unable to use three parts to make the desired broth. ?

4) 2.60

$$\text{World: } 5.6 \times 10^9 \frac{\text{ton CO}_2}{\text{yr}}$$

$$\frac{1}{4}(5.6 \times 10^9 \frac{\text{ton CO}_2}{\text{yr}}) = 1.4 \times 10^9 \frac{\text{ton CO}_2}{\text{yr}}$$

$$x = \frac{4.67 \text{ ton CO}_2}{\text{yr}} \times \frac{10^3 \text{ kg}}{\text{ton}} \times \frac{1}{2.2046 \text{ lb}} \times \frac{1 \text{ lb}}{\text{kg}} = 10288 \frac{\text{lb CO}_2}{\text{yr}}$$



$$\lambda_1 = 1$$

$$\text{C: } 8 =$$

$$\lambda_3$$

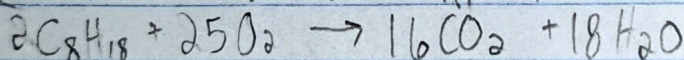
$$\text{H: } 18 =$$

$$2\lambda_4$$

$$\text{O: } 6 = -2\lambda_2 + 2\lambda_3 + \lambda_4$$

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 0 \\ -2 & 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = \begin{bmatrix} 8 \\ 18 \\ 0 \end{bmatrix}$$

$$(\lambda_2 = 2.5 \quad \lambda_3 = 8 \quad \lambda_4 = 9) \times 2$$



$$\text{sg} = .7 = \frac{\rho_{\text{C}_8\text{H}_{18}}}{.997}$$

$$\rho_{\text{C}_8\text{H}_{18}} = .6979 \frac{\text{g}}{\text{mL}}$$

$$\frac{14.231 \text{ g C}_8\text{H}_{18}}{\text{mol}} \times \frac{2 \text{ mol C}_8\text{H}_{18}}{16 \text{ mol CO}_2} \times \frac{1 \text{ mL}}{.6979 \text{ g}} \times \frac{1 \text{ L}}{1000 \text{ mL}} \times \frac{1000 \text{ g}}{\text{kg}} \times \frac{1 \text{ mol CO}_2}{44.01 \text{ g}}$$

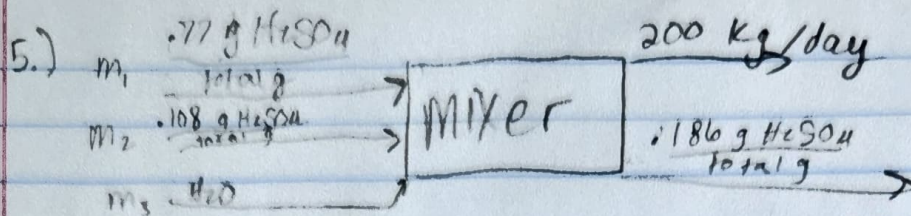
$$= (.465 \frac{\text{L}}{\text{kg CO}_2})^{-1} = 2.15 \frac{\text{kg CO}_2}{\text{L}} \cdot \frac{2.785 \text{ L}}{\text{gal}} = 8.14 \frac{\text{kg CO}_2}{\text{gal}}$$

$$8.14 \frac{\text{kg CO}_2}{\text{gal}} \times \frac{1 \text{ gal}}{3.785 \text{ L}} \times \frac{2.2046 \text{ lb}}{\text{kg}} = .653 \frac{\text{lb}}{\text{mi}} \text{ CO}_2 \quad \checkmark$$

$$.0653 \frac{\text{lb CO}_2}{\text{mi}} \times \frac{14063 \text{ mi}}{\text{yr}} = 9309 \frac{\text{lb CO}_2}{\text{yr}} \text{ per person} \quad \checkmark$$

9303 $\frac{\text{lb}}{\text{yr}}$ person is less than the value on the website 10288 $\frac{\text{lb}}{\text{yr}}$

The website accounted for other vehicles, such as trucks, so it makes sense why our value is lower.



DOF

3 - 2 = 1 underspecified ✓

$$\dot{m}_1 + \dot{m}_2 + \dot{m}_3 = 200$$

$$.77\dot{m}_1 + .108\dot{m}_2 = .186(200) = 37.2$$

Higher cost = 0 to min cost

$$\text{Conc: } \frac{2 \text{¢}}{\text{kg F}} \times \frac{\text{kg F}}{.77 \text{ kg acid}} = 15.987 \text{ ¢/kg acid}$$

$$\text{Diluted: } \frac{2 \text{¢}}{\text{kg R}} \times \frac{\text{kg R}}{.108 \text{ kg acid}} = 18.519 \text{ ¢/kg acid}$$

$$\dot{m}_2 = 0$$

$$\dot{m}_1 + \dot{m}_3 = 200$$

$$.77\dot{m}_1 = 37.2$$

$$\dot{m}_1 = 48.312 \text{ kg/day}$$

$$48.312 + \dot{m}_3 = 200 \quad \dot{m}_3 = 151.688 \text{ kg/day} \quad \checkmark$$

$$\left(\frac{10 \text{¢}}{\text{kg}} \times \frac{48.312 \text{ kg}}{\text{day}} \right) + \left(\frac{2 \text{¢}}{\text{kg}} \times \frac{151.688 \text{ kg}}{\text{day}} \right) = 483.117 \frac{\text{¢}}{\text{day}} \quad \checkmark$$

$$483.117 \frac{\text{¢}}{\text{day}} \times \frac{1 \text{ day}}{200 \text{ kg}} = 2.416 \frac{\text{¢}}{\text{kg}}$$

20/20

good